

UNIT – 5
ASSESSMENT – 3
REPORT

NIHARIKA PREEJESH 192524506

Title:1

System Decomposition and Analysis of an AI Medical Diagnosis System

Objective

To reverse engineer an AI-based medical diagnosis system and identify its major components, data flow, and decision-making process.

Methodology

The system was divided into input, preprocessing, feature extraction, AI decision-making, and output modules. The input symptoms are processed and converted into features. The decision module applies predefined rules to generate a diagnosis.

Result

The reverse-engineered architecture successfully identified the major components and their interactions. The system was able to generate a diagnosis based on the given symptoms.

Title:2

Architecture Reconstruction of a Black-Box Recommendation System

Objective

To reconstruct the architecture of an unknown AI recommendation system by observing its input, processing behaviour, and output.

Methodology

The black-box system was divided into user input, database, recommendation engine, and output modules. The interactions between the modules were reconstructed based on the observed behaviour.

Architecture

Input



Profile Processing



Recommendation Engine



Ranking/Selection



Output

Result

The reconstructed architecture successfully produced recommendations based on the user's interest.

Title:3

Algorithm Identification Through Reverse Engineering

Objective

To identify the underlying decision-making algorithm of a black-box machine learning application.

Methodology

Different input values were provided to the system and the resulting outputs were observed. The decision boundaries were analyzed to identify the underlying algorithm.

Identified Algorithm

The observed decision process resembles a **Decision Tree classifier**, where features are evaluated using threshold-based decisions.

Result

The algorithm and important decision features were successfully identified.

Title 4:

Data Flow and Process Mapping of a Distributed AI Fraud Detection System

Objective

To analyze and map the flow of data through an AI-based fraud detection system.

Methodology

Transaction data is collected, preprocessed, analyzed using risk conditions, and classified into different risk levels.

Result

The system successfully mapped the data processing pipeline and identified high-risk transactions.

Title:5

Improvement and Redesign of an Intelligent Recommendation System

Objective

To reverse engineer an existing intelligent system, identify its limitations, and propose a redesigned architecture for better performance and scalability.

Existing System

The original system performs recommendation generation directly for every user request.

Identified Problems

- Repeated computation
- Slow response for repeated requests
- Limited scalability
- No caching mechanism

Proposed Improvements

The redesigned system introduces a caching layer and separates processing components. In a production system, distributed processing and model-serving components can further improve scalability.

Result

The redesigned implementation avoids repeated computation for the same request by retrieving previously generated results from the cache.